From observed signals to graded Verilog

A model sees only a hidden module's interface and one observed trace. It must write Verilog that passes hidden tests it never saw. This milestone builds the data and the grader underneath.

STEP 1
Sources
5 family generators, 21 design groups
13 local fixture files with a license list

→

STEP 2
Ingest and check
one module, plain ports, safe code, compiles
9 files rejected, each with a reason

→

STEP 3
Dedup and split
exact and comment/whitespace hashes
split by design group, before seeds

→

STEP 4
Simulate
Icarus Verilog, isolated, time limit
reference must match an independent Python model

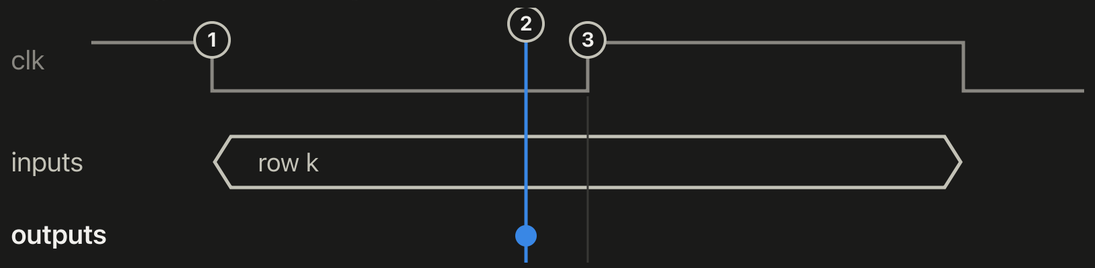
→

Public
interface + observed trace

Target
reference RTL for training

Private
hidden corner cases + 200 random rows

One timing rule for every task, so traces have no races



1 inputs of row k change at the falling edge
2 outputs are sampled 1 ns before the rising edge
3 the rising edge updates the state

66
tasks in the seed-0 build
(62 synthetic, 4 ingested)

222 / 222
deliberate bugs fail
the hidden tests

38
unit tests, all pass
(32 s on a laptop CPU)

What a model gets to see

Real rows from public traces in the seed-0 build: inputs in gray, outputs in color, unknown values before the first reset as a gray band. Each rule is written in plain words, and a separate Python model built from it must agree with the reference RTL on every row.

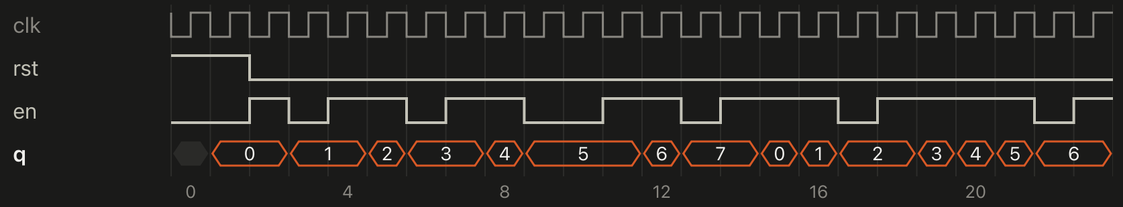
Combinational

y = min(a + b, 15): saturates instead of wrapping · rows 0–15



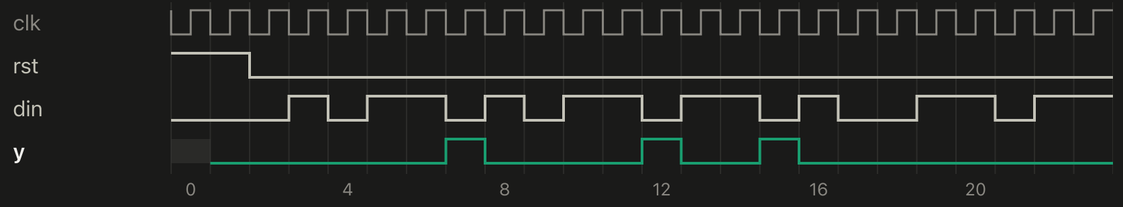
Counter

3-bit counter: reset, count when en, hold, wrap 7 to 0 · rows 0–23



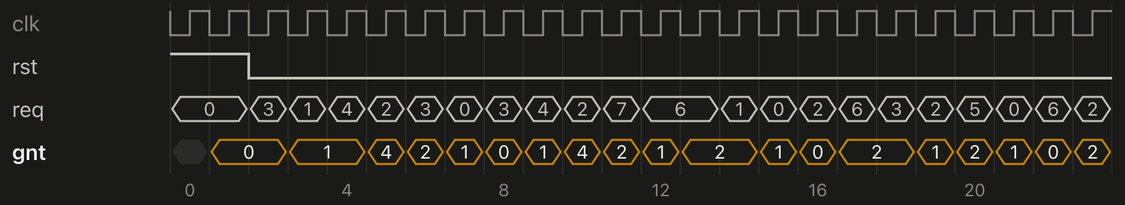
State machine

detects 1011 with overlap; y is high the cycle after the last bit · rows 0–23



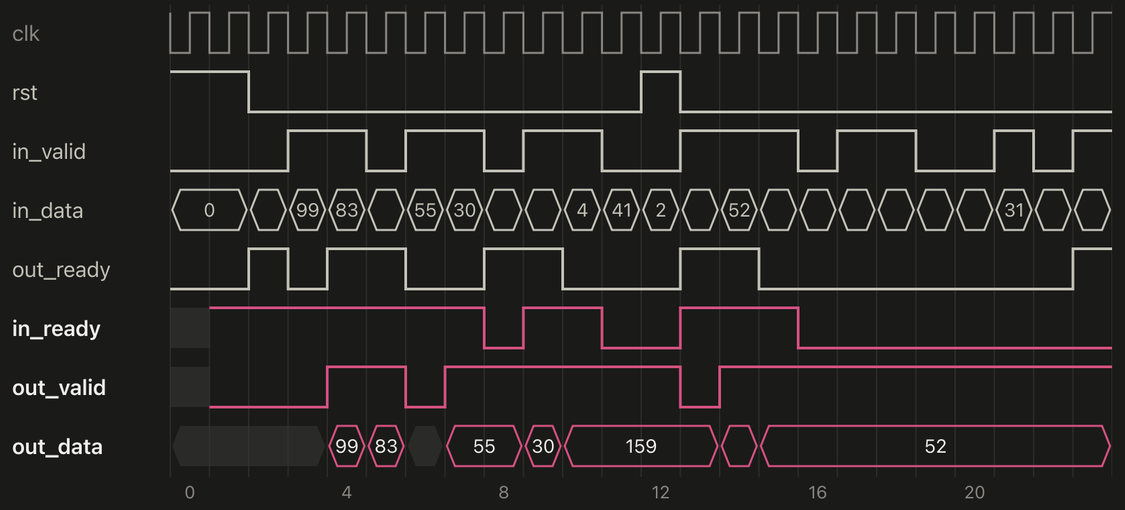
Arbiter

registered one-hot grant: the lowest-numbered request wins (not fair) · rows 0–23



Ready/valid buffer

2-entry buffer: backpressure, in order, data held while stalled · rows 0–23

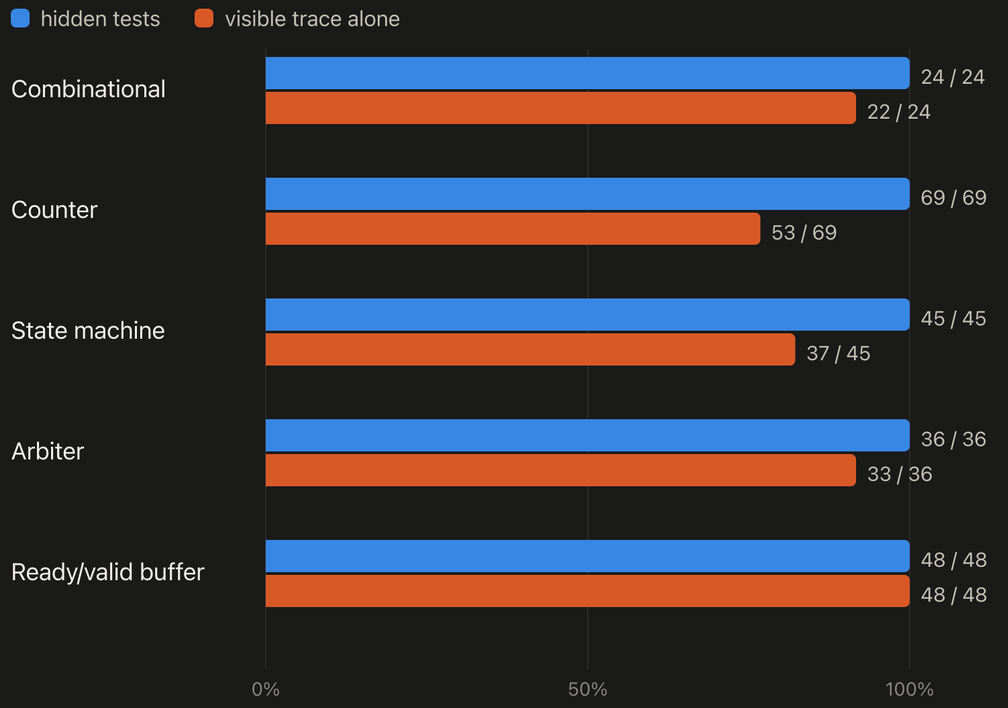


The hidden tests catch the bugs that the trace misses

Every reference passes (62 of 62). Every deliberate bug fails the hidden tests (222 of 222 bug runs). Checked against the visible trace alone, 29 of those bugs would have passed.

Bug runs caught, by family

Bugs include a wrong reset, wrapping one early, a wrong transition, reversed priority, and dropped or duplicated words. Each runs on every task of its family.



Broken candidates get the right status

Six broken versions of a reference per family, 30 of 30 classified as expected.

	syntax error	renamed port	$fatal at run time	early $finish	endless loop	$system call
Combinational	✓ compile_error	✓ interface_error	✓ sim_error	✓ sim_error	✓ timeout	✓ rejected
Counter	✓ compile_error	✓ interface_error	✓ sim_error	✓ sim_error	✓ timeout	✓ rejected
State machine	✓ compile_error	✓ interface_error	✓ sim_error	✓ sim_error	✓ timeout	✓ rejected
Arbiter	✓ compile_error	✓ interface_error	✓ sim_error	✓ sim_error	✓ timeout	✓ rejected
Ready/valid buffer	✓ compile_error	✓ interface_error	✓ sim_error	✓ sim_error	✓ timeout	✓ rejected

What a failure report shows (one real bug run)

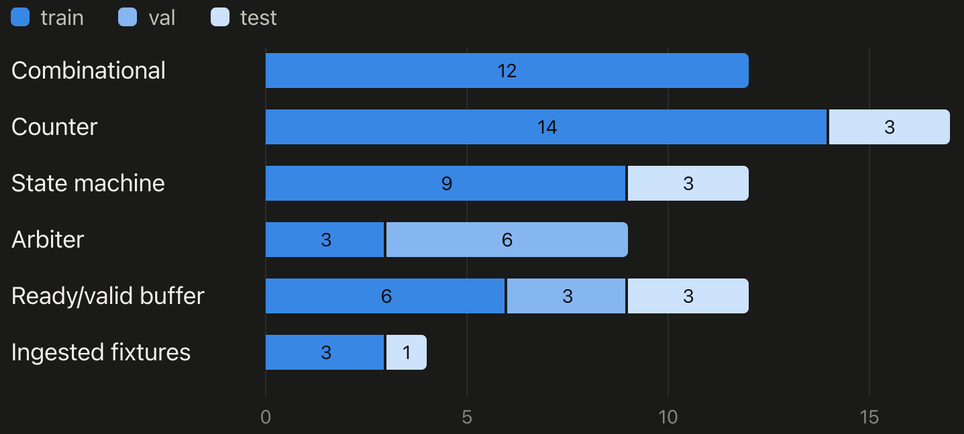
```
candidate: drop_word   status: fail   failing rows: 53 of 242
first failing row: cycle 19
  inputs    in_data=13  in_valid=1  out_ready=1  rst=0
  expected  in_ready=1  out_valid=1  out_data=8
  actual    in_ready=1  out_valid=0  out_data=3
protocol check: cycle 20: delivered 13, expected 8
```

READY FOR BASELINE TRAINING?

The data and the grader are ready; models come next

66 tasks, each with a public trace, a reference target and private hidden tests. Splits are fixed by design group. Model, hardware and budget choices are left to the next stage.

Tasks by family and split



A record (one line of records.jsonl, selected fields)

```
task_id             tee4c6d5546
family              fsm
design_group        fsm:{"pattern": "1011"}
split               train
seed                119217905112749
simulator_version   Icarus Verilog version 13.0 (stable) (v13_0)
reference_status    {"hidden": "pass", "visible": "pass"}
model_check_status  agree
public_trace_path   public/tee4c6d5546/trace.json
private_tests_path  private/tee4c6d5546/hidden.json
```

Status against the milestone

Done	One command builds all five families, with public traces, targets and hidden tests
Done	Harness: 7 distinct statuses, time limits, bounded logs, untrusted code never run
Done	References pass, all bugs fail, broken candidates classified; same seed gives identical bytes
Done	Local ingestion with reasons; exact and normalized dedup; splits by design group
Partial	Dedup is comments and whitespace only: renamed copies are not caught (tested)
Partial	Splits hold new parameter sets of the same 5 templates, not new kinds of designs
Missing	External benchmark ingestion (no license check or download done)
Missing	Synthesis check: a pass is simulation only, not proof that the design synthesizes

Caution: with only 21 synthetic design groups, the split sizes are uneven and some family and split pairs are empty. The seed and the generator are public, so a real held-out evaluation needs a seed that is not published.